Lecture 18 — Matrices as data and structured objects

Mon Oct 5

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Boyd video (optional): Lecture 17 — matrix notation · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 6.1–6.3. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Identify matrix dimensions, rows, columns, and entries.
2. Represent datasets, networks, images, or systems as matrices.
3. Perform matrix addition, scaling, and transposition.
4. Recognize zero, identity, diagonal, triangular, symmetric, and sparse matrices.
5. Explain one practical consequence of matrix structure.

Warm-up

Let

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 1 \end{bmatrix}.$$

(a) What are the dimensions of A? Is it tall, wide, or square?

(b) What is A_{23}? What is the second column of A as a vector?

Fill in

An $m \times n$ *matrix* is a rectangular array with _____ rows and _____ columns. The entry in row i and column j is written A_{ij} (Boyd: indices start at 1).

We can view A as its n **columns** (each an m-vector) side by side, or as its m **rows** (each an n-row-vector) stacked.

Examples of "matrix as data":

- An image: pixel values in a grid.
- A table: locations × days, or customers × products.
- A **relation / directed graph**: adjacency matrix with $A_{ij} = 1$ when there is an edge from j to i (Boyd's convention).

Transpose. $(A^\mathsf{T})_{ij} =$ _____. If A is $m \times n$, then A^T is _____.

Addition and scaling. $A + B$ and βA are entrywise (same size required for addition).

Special matrices (square, unless noted):

Name	Pattern
Zero 0	all entries _____
Identity I	1s on the diagonal, 0s elsewhere
Diagonal	off-diagonal entries are _____
Upper triangular	zeros _____ the diagonal
Symmetric	$A^\mathsf{T} =$ _____
Sparse	_____ of the entries are nonzero

Practical consequence: structure (sparsity, triangularity, symmetry) often means cheaper storage or faster algorithms — we take advantage of the pattern instead of treating every entry as generic.

Worked in class

W1. Dimensions and pieces. Let

$$B = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 4 & 5 \\ 0 & 1 & 1 \end{bmatrix}.$$

Give B_{31}, the third row, and the first column. Is B square?

W2. Data matrix. Three weather stations record (high °F, low °F) for one day. Station 1: $(62, 48)$; station 2: $(58, 45)$; station 3: $(65, 50)$. Write a 2×3 matrix W whose columns are the station readings. What does W_{12} mean?

W3. Arithmetic. With

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix},$$

compute $A + C$, $2A$, and A^{T}.

W4. Structure check. For each matrix, name every special type that applies (zero / identity / diagonal / triangular / symmetric / sparse — sparse is judgment call for tiny examples).

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

W5. Why structure matters. An $n \times n$ identity matrix has how many nonzero entries? Density $nnz/(n^2)$? Why might software store only the nonzeros of a sparse customer-purchase matrix?

Your turn

Y1. Let

$$M = \begin{bmatrix} 0 & 2 & 1 \\ -1 & 3 & 4 \end{bmatrix}.$$

Dimensions? M_{21}? Second column? Is M tall or wide?

Y2. A tiny directed graph on nodes $\{1, 2, 3\}$ has edges $2 \to 1$, $1 \to 3$, and $3 \to 2$. Using Boyd's convention ($A_{ij} = 1$ means edge from j to i), write the 3×3 adjacency matrix.

Y3. With $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$, compute $A + A^{\mathsf{T}}$. Is the result symmetric? Why?

Y4. True or false, with a one-line reason: every diagonal matrix is symmetric.

Answers (Your turn)

Y1. 2×3; wide. $M_{21} = -1$. Second column $(2, 3)$.

Y2.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

(edge $2 \to 1$ puts a 1 in row 1, column 2; $1 \to 3$ puts a 1 in row 3, column 1; $3 \to 2$ puts a 1 in row 2, column 3).

Y3. $A + A^{\mathsf{T}} = \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix}$, which equals its transpose, so yes — sum of a matrix and its transpose is always symmetric.

Y4. **True.** Off-diagonal entries are already $0 = 0$, so $A^{\mathsf{T}} = A$.